

AYMANE AIT DADS

Forward Deployed Engineer | Data Pipeline Design · ML Workflow Automation · Applied AI Infrastructure
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PROFESSIONAL SUMMARY

Ranked 17th out of 3,000+ teams in the NVIDIA Nemotron Reasoning Challenge (score 0.86/1.0, \$106K+ prize pool) by engineering an end-to-end LoRA SFT pipeline on a 30B-parameter model with custom data verification infrastructure. Reduced manual analysis time by 60% at Orange Maroc by building automated feature engineering and clustering pipelines over 100,000+ fiber-optic network records, translating model outputs directly into production maintenance recommendations. Brings strong Python proficiency, hands-on data wrangling and preprocessing experience, and deep familiarity with ML workflows including dataset curation, evaluation pipelines, and ablation studies - exactly the profile Isidor needs to build reliable AI evaluation infrastructure. Aims to design scalable data pipelines and model-in-the-loop quality control systems that push the frontier of measurable, reproducible AI evaluation for leading research labs and enterprise teams.

CORE COMPETENCIES

Data Pipelines | Python Automation | Data Wrangling & Preprocessing | Machine Learning Workflows | Evaluation Pipelines | Applied Mathematical Reasoning | Model-in-the-Loop Systems | Dataset Curation

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Designed an **end-to-end data pipeline** ingesting 100,000+ fiber-optic network records (upload speed, latency, jitter) with automated feature engineering, reducing manual analysis time by 60%.
- Built a clustering pipeline mapping customer performance profiles to **5 commercial service tiers** using K-Means, with output adopted directly by the network optimization team for production decisions.
- Developed Random Forest classification models to assess network quality across thousands of nodes, applying rigorous **data wrangling and preprocessing** to raw telemetry data before model ingestion.
- Delivered stakeholder presentations translating complex model outputs into actionable maintenance recommendations, demonstrating strong **communication skills** bridging technical findings and business teams.

PROJECTS

NVIDIA Nemotron Model Reasoning Challenge

Kaggle Competition · In Progress (June 2026) · Ranked 17th / 3,000+ Teams

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, \$106K+ prize pool) in the ongoing NVIDIA Nemotron Reasoning Challenge by fine-tuning a 30B-parameter Mamba-Transformer with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context).
- Ran **controlled ablation experiments** across LR schedules, packing strategies, and synthetic augmentation - discovering reasoning-style consistency outweighs data volume in LLM SFT; built custom cipher parsers and 8-bit operation solvers producing verified CoT datasets with zero boxed mismatches.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Kaggle GPU

AI-Generated Image Detection

EURECOM ImSecu + NTIRE 2026 @ CVPR · 2025 · Ranked 1st in Class

- Ranked **1st in EURECOM class** (private Kaggle leaderboard, 0.791 AUC) by fine-tuning CLIP ViT-L/14 (428M params) with a custom MLP head, dual learning rates, and 10-view TTA across 250,000 training images from 25+ generator types.
- Submitted independently to **NTIRE 2026 @ CVPR CodaBench** international benchmark; discovered that 1-epoch training (0.793 AUC) outperforms 4-epoch training (0.767 AUC), demonstrating OOD generalization through disciplined evaluation pipeline design.

PyTorch · OpenCLIP · ViT · FP16 Mixed Precision · Kaggle GPU

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Statistics, Computer Vision, NLP, Cloud Computing, Image Security, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

ML Frameworks & Engineering: PyTorch | Scikit-learn | Hugging Face Transformers | Unsloth | TRL | PEFT | vLLM

Data & Automation: Python | Pandas | NumPy | SQL | feature engineering | EDA | Power BI

LLM & AI Systems: LoRA | SFTTrainer | chain-of-thought | RAG | LangChain | prompt engineering | agentic pipelines